

Movie Review Sentiment Analysis - Viva + Practical Preparation

This PDF contains:

- Deep Learning and NLP concepts
- Line-by-line explanation
- Dataset commands
- Pandas DataFrame commands
- Important viva questions and answers
- CNN + LSTM model explanation

1. What is Sentiment Analysis?

Sentiment Analysis means identifying whether a text is positive or negative.

2. What is NLP?

NLP stands for Natural Language Processing. It helps computers understand human language.

3. What is Deep Learning?

Deep Learning uses neural networks with many layers to learn patterns from data.

4. What is IMDB Dataset?

IMDB dataset contains movie reviews labeled as positive or negative.

5. What is Tokenization?

Tokenization converts words into numbers.

6. What is Padding?

Padding makes all reviews equal length.

7. What is Embedding Layer?

Embedding converts words into dense vectors.

8. What is CNN?

CNN extracts important features from text.

9. What is LSTM?

LSTM remembers sequence information in text.

10. What is Dropout?

Dropout prevents overfitting.

11. What is Sigmoid Function?

Sigmoid gives output between 0 and 1.

12. What is Epoch?

One complete training cycle over dataset.

13. What is Batch Size?

Number of samples processed together.

Important Dataset Commands

Command	Purpose
df.head()	Shows first 5 rows
df.tail()	Shows last 5 rows
df.shape	Shows rows and columns
len(df)	Returns total rows
df.info()	Shows dataset information
df.describe()	Statistical summary
df.columns	Shows column names
df.dtypes	Shows data types
df.isnull()	Checks null values
df.isnull().sum()	Counts null values
df.sample(5)	Displays random rows
df['column'].unique()	Shows unique values
df['column'].value_counts()	Counts occurrences

Common Viva Questions and Answers

Q: What is Sentiment Analysis?

A: It identifies whether text is positive or negative.

Q: Why use NLP?

A: NLP helps computers understand human language.

Q: What is CNN used for?

A: CNN extracts features from text.

Q: What is LSTM used for?

A: LSTM remembers sequence information.

Q: Why use padding?

A: Neural networks require equal input length.

Q: What is overfitting?

A: When model performs well on training data but poorly on new data.

Q: Why use Dropout?

A: To prevent overfitting.

Q: What is sigmoid activation?

A: It converts output into probability between 0 and 1.

Q: What is accuracy?

A: Percentage of correct predictions.

Q: What is binary classification?

A: Classification into two classes.

Basic Python and Dataset Commands

```
print(X_train.shape)
print(len(X_train))
print(X_train[0])
print(y_train[0])
print(type(X_train))
print(np.max(X_train))
print(np.min(X_train))
model.summary()
model.predict(X_test[:1])
```

Short Viva Summary

This project performs sentiment analysis using CNN and LSTM deep learning models on IMDB movie reviews. The reviews are converted into numerical sequences using tokenization and padding. CNN extracts features while LSTM learns sequence dependencies. Finally, sigmoid activation classifies reviews as positive or negative.